

Teichmuller theory Problem set 3

1. Let $X = \mathbb{C}/L$ where L is a lattice in $\mathbb{C}$, and let $\pi : \mathbb{C} \rightarrow X$ be the universal covering map. By a line segment in X we mean a curve $\gamma : I \subset \mathbb{R} \rightarrow X$ in X such that any lift of γ is a straight line in $\mathbb{C}$. Let $g : X \rightarrow X$ be a QC map which is homotopic to the identity.

a) Show that there exists a constant $M > 0$ such that $l(g(\gamma)) \geq l(\gamma) - 2M$ for all line segments γ in X (here $l(\gamma)$ denotes the length of a curve γ with respect to the canonical flat metric on X).

b) Let $\{z_\alpha\}_{\alpha \in I}$ denote the collection of charts on X given by local inverses of the covering map π . Let λ_g be the measurable function on X defined by $\lambda_g(p) = |\frac{\partial}{\partial x}(z_\beta \circ g \circ z_\alpha^{-1})(z_\alpha(p))|$ for a.e. $p \in X$, where z_α, z_β are charts near p and $g(p)$ respectively. Show that

$$\frac{1}{\text{Area}(X)} \int_X \lambda_g \, d\text{vol}_X \geq 1$$

(here $d\text{vol}_X$ and $\text{Area}(X)$ denote the area form on X and the area of X with respect to the canonical flat metric on X).

2. Let X, Y be compact Riemann surfaces of genus $g \geq 2$ and let $f : X \rightarrow Y$ be a QC map. Show that there exists a QC map $f_0 : X \rightarrow Y$ such that f_0 is homotopic to f and such that $K(f_0) \leq K(h)$ for all QC maps $h : X \rightarrow Y$ which are homotopic to f .

3. For $g \geq 2$, let $\mathcal{T}_g$ denote the Teichmuller space of Riemann surfaces of genus g , equipped with the Teichmuller metric. Suppose $[X_n, f_n] \rightarrow [X, f]$ in $\mathcal{T}_g$ as $n \rightarrow \infty$. Show that there are marked Riemann surfaces $(Y_n, g_n) \sim (X_n, f_n)$ and QC maps $\phi_n : X \rightarrow Y_n$ such that $\phi_n \circ f$ is homotopic to g_n , and such that ϕ_n has a lift $\tilde{\phi}_n : \mathbb{D} \rightarrow \mathbb{D}$ such that $\tilde{\phi}_n \rightarrow id, \tilde{\phi}_n^{-1} \rightarrow id$ uniformly on compacts in $\mathbb{D}$ as $n \rightarrow \infty$.

4. For $g \geq 2$, let $\mathcal{T}_g$ denote the Teichmuller space of Riemann surfaces of genus g , equipped with the Teichmuller metric. Show that $\mathcal{T}_g$ is a complete metric space.

5. Show that the mapping class group Mod_g acts by isometries on the Teichmuller space $\mathcal{T}_g$.

6. Let (X, f) be a marked Riemann surface of genus $g \geq 2$. Show that the stabilizer of $[X, f] \in \mathcal{T}_g$ in the mapping class group Mod_g is isomorphic to $\text{Aut}(X)$.

7. Let X be a compact Riemann surface of genus $g \geq 2$.

a) Show that any holomorphic quadratic differential ϕ on X which is not identically zero has $4g - 4$ zeroes counted with multiplicity.

b) Show that the $\mathbb{C}$ -vector space $Q(X)$ of holomorphic quadratic differentials on X has dimension $3g - 3$.

8. Let $g \geq 2$ and let $[X, h] \in \mathcal{T}_g$. Let $\phi \in Q(X) - \{0\}$, let $a_t \cdot (X, \phi) = (X_t, \phi_t)$ for $t \in \mathbb{R}$ where $a_t \in \text{SL}(2, \mathbb{R})$ is the diagonal matrix with diagonal entries $e^{t/2}, e^{-t/2}$, and let $f_t : X \rightarrow X_t$ be the corresponding Teichmuller mapping with initial and terminal differentials ϕ and ϕ_t respectively such that $K(f_t) = e^t$.

a) Show that the curve $\gamma_0(t) = [X_t, f_t \circ h] \in \mathcal{T}_g$ is a geodesic in $\mathcal{T}_g$.

b) Show that if $\gamma : [0, T] \rightarrow \mathcal{T}_g$ is a geodesic such that $\gamma(0) = \gamma_0(0), \gamma(T) = \gamma_0(T)$, then $\gamma(t) = \gamma_0(t)$ for all $t \in [0, T]$.

9. Let X, Y be compact Riemann surfaces of genus $g \geq 2$ and let $\phi \in Q(X) - \{0\}, \psi \in Q(Y) - \{0\}$. Let $f_0 : X \rightarrow Y$ be a QC map such that there is an $A \in \mathrm{SL}(2, \mathbb{R})$ such that for any natural parameters ξ, η of ϕ, ψ respectively, we have $\eta \circ f_0 \circ \xi^{-1} = \pm A + c$ for some constant c (depending on ξ, η). Show that if $f : X \rightarrow Y$ is a QC map homotopic to f_0 then $K(f) \geq K(f_0)$, with equality if and only if $f = f_0$.